

Q1.

AES operates on a block of 128 bits arranged as a 4x4 state matrix (each element is 1 byte).

Suppose the following plaintext (in hexadecimal) is the input block:

$$\text{Plaintext Block} = \begin{bmatrix} 32 & 88 & 31 & e0 \\ 43 & 5a & 31 & 37 \\ f6 & 30 & 98 & 07 \\ a8 & 8d & a2 & 34 \end{bmatrix}$$

and the first round key (in hex) is:

$$\text{Round Key} = \begin{bmatrix} 2b & 28 & ab & 09 \\ 7e & ae & f7 & cf \\ 15 & d2 & 15 & 4f \\ 16 & a6 & 88 & 3c \end{bmatrix}$$

Perform the **AddRoundKey** step of AES (bitwise XOR of plaintext block with round key). Show the resulting state matrix.

Answer :

$$= \begin{bmatrix} 19 & A0 & 9A & E9 \\ 3D & F4 & C6 & F8 \\ E3 & E2 & 8D & 4E \\ BE & 2B & 2A & 08 \end{bmatrix}$$

$$\begin{array}{l} 32 = 0011 \ 0010 \\ \oplus \\ 2b = 0010 \ 1011 \\ \hline = 0001 \ 1001 \\ \text{1} \quad \text{9} \end{array}$$

$$\begin{array}{l} 43 = 0100 \ 0011 \\ \oplus \\ 7e = 0111 \ 1110 \\ \hline 0011 \ 1101 \\ \text{3} \quad \text{D} \end{array}$$

$$\begin{array}{l} f6 = 1111 \ 0110 \\ \oplus \\ 15 = 0001 \ 0101 \\ \hline 1110 \ 0011 \\ \text{E} \quad \text{3} \end{array}$$

$$\begin{array}{l} a8 = 1010 \ 1000 \\ \oplus \\ 16 = 0001 \ 0110 \\ \hline 1011 \ 1110 \\ \text{B} \quad \text{E} \end{array}$$

$$\begin{array}{l} 88 = 1000 \ 1000 \\ \oplus \\ 28 = 0010 \ 1000 \\ \hline 1010 \ 0000 \\ \text{A} \quad \text{0} \end{array}$$

$$\begin{array}{l} 5a = 0101 \ 1010 \\ \oplus \\ ae = 1010 \ 1110 \\ \hline 1111 \ 0100 \\ \text{F} \quad \text{4} \end{array}$$

$$\begin{array}{l} 30 = 0011 \ 0000 \\ \oplus \\ d2 = 1101 \ 0010 \\ \hline 1110 \ 0010 \\ \text{E} \quad \text{2} \end{array}$$

$$\begin{array}{l} 8d = 1000 \ 1101 \\ \oplus \\ 96 = 1010 \ 0110 \\ \hline 0010 \ 1011 \\ \text{2} \quad \text{6} \end{array}$$

$$\begin{array}{l} 31 = 0011 \ 0001 \\ \oplus \\ ab = 1010 \ 1011 \\ \hline 1001 \ 1010 \\ \text{9} \quad \text{A} \end{array}$$

$$\begin{array}{l} 31 = 0011 \ 0001 \\ \oplus \\ f7 = 1111 \ 0111 \\ \hline 1100 \ 0110 \\ \text{C} \quad \text{6} \end{array}$$

$$\begin{array}{l} 98 = 1001 \ 1000 \\ \oplus \\ 15 = 0001 \ 0101 \\ \hline 1000 \ 1101 \\ \text{8} \quad \text{D} \end{array}$$

$$\begin{array}{l} a2 = 1010 \ 0010 \\ \oplus \\ 88 = 1000 \ 1000 \\ \hline 0010 \ 1010 \\ \text{2} \quad \text{A} \end{array}$$

$$\begin{array}{l} e0 = 1110 \ 0000 \\ \oplus \\ 09 = 0000 \ 1001 \\ \hline 1110 \ 1001 \\ \text{E} \quad \text{9} \end{array}$$

$$\begin{array}{l} 37 = 0011 \ 0111 \\ \oplus \\ cf = 1100 \ 1111 \\ \hline 1111 \ 1000 \\ \text{F} \quad \text{8} \end{array}$$

$$\begin{array}{l} 07 = 0000 \ 0111 \\ \oplus \\ 4f = 0100 \ 1111 \\ \hline 0100 \ 1000 \\ \text{4} \quad \text{8} \end{array}$$

$$\begin{array}{l} 34 = 0011 \ 0100 \\ \oplus \\ 3c = 0011 \ 1100 \\ \hline 0000 \ 1000 \\ \text{0} \quad \text{8} \end{array}$$

Q2:

You have the AES state (hex) right *before* SubBytes:

$$\begin{bmatrix} 19 & a0 & 9a & e9 \\ 3d & f4 & c6 & f8 \\ e3 & e2 & 8d & 48 \\ be & 2b & 2a & 08 \end{bmatrix}$$

Task. Apply SubBytes (use the standard AES S-box). Show the resulting state.

Table 5.2 AES S-Boxes

|   |   | y  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|   |   | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | A  | B  | C  | D  | E  | F  |
| x | 0 | 63 | 7C | 77 | 7B | F2 | 6B | 6F | C5 | 30 | 01 | 67 | 2B | FE | D7 | AB | 76 |
|   | 1 | CA | 82 | C9 | 7D | FA | 59 | 47 | F0 | AD | D4 | A2 | AF | 9C | A4 | 72 | C0 |
|   | 2 | B7 | FD | 93 | 26 | 36 | 3F | F7 | CC | 34 | A5 | E5 | F1 | 71 | D8 | 31 | 15 |
|   | 3 | 04 | C7 | 23 | C3 | 18 | 96 | 05 | 9A | 07 | 12 | 80 | E2 | EB | 27 | B2 | 75 |
|   | 4 | 09 | 83 | 2C | 1A | 1B | 6E | 5A | A0 | 52 | 3B | D6 | B3 | 29 | E3 | 2F | 84 |
|   | 5 | 53 | D1 | 00 | ED | 20 | FC | B1 | 5B | 6A | CB | BE | 39 | 4A | 4C | 58 | CF |
|   | 6 | D0 | EF | AA | FB | 43 | 4D | 33 | 85 | 45 | F9 | 02 | 7F | 50 | 3C | 9F | A8 |
|   | 7 | 51 | A3 | 40 | 8F | 92 | 9D | 38 | F5 | BC | B6 | DA | 21 | 10 | FF | F3 | D2 |
|   | 8 | CD | 0C | 13 | EC | 5F | 97 | 44 | 17 | C4 | A7 | 7E | 3D | 64 | 5D | 19 | 73 |
|   | 9 | 60 | 81 | 4F | DC | 22 | 2A | 90 | 88 | 46 | EE | B8 | 14 | DE | 5E | 0B | DB |
|   | A | E0 | 32 | 3A | 0A | 49 | 06 | 24 | 5C | C2 | D3 | AC | 62 | 91 | 95 | E4 | 79 |
|   | B | E7 | C8 | 37 | 6D | 8D | D5 | 4E | A9 | 6C | 56 | F4 | EA | 65 | 7A | AE | 08 |
|   | C | BA | 78 | 25 | 2E | 1C | A6 | B4 | C6 | E8 | DD | 74 | 1F | 4B | BD | 8B | 8A |
|   | D | 70 | 3E | B5 | 66 | 48 | 03 | F6 | 0E | 61 | 35 | 57 | B9 | 86 | C1 | 1D | 9E |
|   | E | E1 | F8 | 98 | 11 | 69 | D9 | 8E | 94 | 9B | 1E | 87 | E9 | CE | 55 | 28 | DF |
|   | F | 8C | A1 | 89 | 0D | BF | E6 | 42 | 68 | 41 | 99 | 2D | 0F | B0 | 54 | BB | 16 |

(a) S-box

Answer :

|    |    |    |    |
|----|----|----|----|
| D4 | E0 | B8 | 1E |
| 27 | BF | B4 | 41 |
| 11 | 98 | 5D | 52 |
| AE | F1 | E5 | 30 |

Q3: Give me the output of mixcolumns for only one column in the state:

Consider one column after SubBytes (top → bottom bytes in hex):

$$c = \begin{bmatrix} d4 \\ bf \\ 5d \\ 30 \end{bmatrix}$$

AES MixColumns multiplies each column by the fixed matrix over  $GF(2^8)$ :

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \cdot \begin{bmatrix} d4 \\ bf \\ 5d \\ 30 \end{bmatrix}$$

Answer :

First element

$$d4 * 02 \oplus bf * 03 \oplus 5d * 01 \oplus 30 * 01$$

$$(d4 * 02) \Rightarrow \begin{array}{r} 1101 \ 0100 * 2 \\ 1010 \ 1000 \\ 0001 \ 1011 \oplus \\ \hline 1011 \ 0011 \end{array}$$

$$(bf * 03) \Rightarrow bf * 02 \oplus bf * 01$$

$$(bf * 01) \Rightarrow 1011 \ 1111$$

$$(bf * 02) \Rightarrow \begin{array}{r} 1011 \ 1111 * 2 \\ 0111 \ 1110 \\ 0001 \ 1011 \oplus \\ \hline 0110 \ 0101 \end{array}$$

$$(5d * 01) \Rightarrow 0101 \ 1101$$

$$(30 * 01) \Rightarrow 0011 \ 0000$$

$$\text{First element} = 10110011 \oplus 10111111 \oplus 01100101 \oplus 01011101 \oplus 00110000$$

$$= 0000 \ 0100 = 04$$

Second element

$$(d4 * 01) \oplus bf * 02 \oplus 5d * 03 \oplus 30 * 01$$

$$(d4 * 01) \Rightarrow 1101 \ 0100$$

$$(bf * 02) \Rightarrow \begin{array}{r} 1011 \ 1111 * 2 \\ 0111 \ 1110 \\ 0001 \ 1011 \oplus \\ \hline 0110 \ 0101 \end{array}$$

$$(5d * 03) \Rightarrow 5d * 02 \oplus 5d * 01$$

$$(5d * 01) \Rightarrow 0101 \ 1101$$

$$(5d * 02) \Rightarrow \begin{array}{r} 0101 \ 1101 * 2 \\ 1011 \ 1010 \\ \hline 1011 \ 1010 \end{array}$$

$$(30 * 01) \Rightarrow 0011 \ 0000$$

$$\text{Second element} = 1101 \ 0100 \oplus 0110 \ 0101 \oplus 0101 \ 1101 \oplus 1011 \ 1010 \oplus 0011 \ 0000 = 0110 \ 0110 = 66$$

Third element

$$d4 * 01 \oplus bf * 01 \oplus 5d * 02 \oplus 30 * 03$$

$$(d4 * 01) \Rightarrow 1101 \ 0100$$

$$(bf * 01) \Rightarrow 1011 \ 1111$$

$$(5d * 02) \Rightarrow \begin{array}{r} 0101 \ 1101 * 2 \\ 1011 \ 1010 \\ \hline 1011 \ 1010 \end{array}$$

$$(30 * 03) \Rightarrow (30 * 02) \oplus (30 * 01)$$

$$(30 * 01) \Rightarrow 0011 \ 0000$$

$$(30 * 02) \Rightarrow \begin{array}{r} 0011 \ 0000 * 2 \\ 0110 \ 0000 \\ \hline 0110 \ 0000 \end{array}$$

$$\text{Third element} = 1101 \ 0100 \oplus 1011 \ 1111 \oplus 1011 \ 1010 \oplus 0011 \ 0000 \oplus 0110 \ 0000 = 1000 \ 0001 = 81$$

Fourth element

$$d4 * 03 \oplus bf * 01 \oplus 5d * 01 \oplus 30 * 02$$

$$(d4 * 03) \Rightarrow (d4 * 02) \oplus (d4 * 01)$$

$$(d4 * 01) \Rightarrow 1101 \ 0100$$

$$(d4 * 02) \Rightarrow \begin{array}{r} 1101 \ 0100 * 2 \\ 1010 \ 1000 \\ 0001 \ 1011 \oplus \\ \hline 1011 \ 0011 \end{array}$$

$$(bf * 01) \Rightarrow 1011 \ 1111$$

$$(5d * 01) \Rightarrow 0101 \ 1101$$

$$(30 * 02) \Rightarrow \begin{array}{r} 0011 \ 0000 * 2 \\ 0110 \ 0000 \\ \hline 0110 \ 0000 \end{array}$$

$$\text{Fourth element} = 1101 \ 0100 \oplus 1011 \ 0011 \oplus 1011 \ 1111 \oplus 0101 \ 1101 \oplus 0110 \ 0000 = 1110 \ 0101 = E5$$

Result =

|    |
|----|
| 04 |
| 66 |
| 81 |
| E5 |

## Q4) AES Key Expansion (Key Schedule)

AES-128 uses a 128-bit key (16 bytes), expanded into 11 round keys (each 16 bytes).

Suppose the initial cipher key (in hex) is:

$$\text{Key} = \begin{bmatrix} 2b & 7e & 15 & 16 \\ 28 & ae & d2 & a6 \\ ab & f7 & 97 & 75 \\ 46 & 20 & 64 & 3b \end{bmatrix}$$

**Task.** Perform the first step of key expansion (generation of round key 1).

That means:

1. Split key into 4 words:  $W[0], W[1], W[2], W[3]$ .
2. Compute  $W[4]$  using the AES schedule formula:

$$W[4] = W[0] \oplus \text{SubWord}(\text{RotWord}(W[3])) \oplus \text{Rcon}[1]$$

3. Then compute  $W[5], W[6], W[7]$ .

Finally, write Round Key 1.

$$\text{Rcon}[1] = \{01, 00, 00, 00\}$$

Table 5.2 AES S-Boxes

|   |   | y  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
|   |   | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | A  | B  | C  | D  | E  | F  |
| x | 0 | 63 | 7C | 77 | 7B | F2 | 6B | 6F | C5 | 30 | 01 | 67 | 2B | FE | D7 | AB | 76 |
|   | 1 | CA | 82 | C9 | 7D | FA | 59 | 47 | F0 | AD | D4 | A2 | AF | 9C | A4 | 72 | C0 |
|   | 2 | B7 | FD | 93 | 26 | 36 | 3F | F7 | CC | 34 | A5 | E5 | F1 | 71 | D8 | 31 | 15 |
|   | 3 | 04 | C7 | 23 | C3 | 18 | 96 | 05 | 9A | 07 | 12 | 80 | E2 | EB | 27 | B2 | 75 |
|   | 4 | 09 | 83 | 2C | 1A | 1B | 6E | 5A | A0 | 52 | 3B | D6 | B3 | 29 | E3 | 2F | 84 |
|   | 5 | 53 | D1 | 00 | ED | 20 | FC | B1 | 5B | 6A | CB | BE | 39 | 4A | 4C | 58 | CF |
|   | 6 | D0 | EF | AA | FB | 43 | 4D | 33 | 85 | 45 | F9 | 02 | 7F | 50 | 3C | 9F | A8 |
|   | 7 | 51 | A3 | 40 | 8F | 92 | 9D | 38 | F5 | BC | B6 | DA | 21 | 10 | FF | F3 | D2 |
|   | 8 | CD | 0C | 13 | EC | 5F | 97 | 44 | 17 | C4 | A7 | 7E | 3D | 64 | 5D | 19 | 73 |
|   | 9 | 60 | 81 | 4F | DC | 22 | 2A | 90 | 88 | 46 | EE | B8 | 14 | DE | 5E | 0B | DB |
|   | A | F0 | 32 | 3A | 0A | 49 | 06 | 24 | 5C | C2 | D3 | AC | 62 | 91 | 95 | E4 | 79 |
|   | B | E7 | C8 | 37 | 6D | 8D | D5 | 4E | A9 | 6C | 56 | F4 | EA | 65 | 7A | AE | 08 |
|   | C | BA | 78 | 25 | 2E | 1C | A6 | B4 | C6 | E8 | DD | 74 | 1F | 4B | BD | 8B | 8A |
|   | D | 70 | 3E | B5 | 66 | 48 | 03 | F6 | 0E | 61 | 35 | 57 | B9 | 86 | C1 | 1D | 9E |
|   | E | F1 | F8 | 98 | 11 | 69 | D9 | 8E | 94 | 9B | 1E | 87 | F9 | CE | 55 | 28 | DF |
|   | F | 8C | A1 | 89 | 0D | BF | E6 | 42 | 68 | 41 | 99 | 2D | 0F | B0 | 54 | BB | 16 |

(a) S-box

1)

$$W_0 = [2b, 28, ab, 46]$$

$$W_1 = [7e, ae, f7, 20]$$

$$W_2 = [15, d2, 97, 64]$$

$$W_3 = [16, a6, 75, 3b]$$

$$2) W[4] = W[0] \oplus \text{SubWord}(\text{RotWord}(W[3])) \oplus \text{Rcon}[1]$$

$$1 - \text{RotWord}(W[3]):$$

$$\text{RotWord}([16, a6, 75, 3b]) = [a6, 75, 3b, 16]$$

$$2 - \text{SubWord}$$

Apply AES S-Box to each byte. Using the AES S-Box.

$$\text{SubWord}([a6, 75, 3b, 16]) = [24, 9d, e2, 47]$$

$$W[4] = W[0] \oplus \text{SubWord}(\text{RotWord}(W[3])) \oplus \text{Rcon}[1]$$

$$= [2b, 28, ab, 46] \oplus [24, 9d, e2, 47] \oplus [01, 00, 00, 00]$$

$$= [0e, b5, 49, 01]$$

$$3) W[5] = W[4] \oplus W[1] = [0e, b5, 49, 01] \oplus [7e, ae, f7, 20] = [70, 1b, be, 21]$$

$$W[6] = W[5] \oplus W[2] = [70, 1b, be, 21] \oplus [15, d2, 97, 64] = [65, c9, 29, 45]$$

$$W[7] = W[6] \oplus W[3] = [65, c9, 29, 45] \oplus [16, a6, 75, 3b] = [73, 6f, 5c, 7e]$$

K1 =

|    |    |    |    |
|----|----|----|----|
| 0e | 70 | 65 | 73 |
| b5 | 1b | c9 | 6f |
| 49 | be | 29 | 5c |
| 01 | 21 | 45 | 7e |